

Chapter 4, Section 5

James Lee

June 2, 2026

1. Show that a hyperelliptic curve can never be a complete intersection in any projective space.

Proof. Combine (Ex. 3.3) and (5.2). □

2. If X is a curve of genus ≥ 2 over a field of characteristic 0, show that the group $\text{Aut } X$ of automorphisms of X is finite.

Proof. Suppose X is hyperelliptic, and let $f : X \rightarrow \mathbf{P}^1$ be a 2-fold covering $f : X \rightarrow \mathbf{P}^1$ given by the unique g_2^1 . The ramification locus consists of $2g + 2$ distinct points. After a suitable automorphism of $\mathbf{P}^1$, we can label them $0, 1, \infty, \beta_1, \dots, \beta_{2g-1}$. Let σ be an automorphism of X . Then $f \circ \sigma$ gives a g_2^1 , so by uniqueness of g_2^1 , there exists an automorphism τ of $\mathbf{P}^1$ such that the following diagram commutes

$$\begin{array}{ccc} X & \xrightarrow{\sigma} & X \\ \downarrow & & \downarrow \\ \mathbf{P}^1 & \xrightarrow{\tau} & \mathbf{P}^1 \end{array} .$$

Thus, σ permutes the ramification points of f which induces a group homomorphism $\text{Aut } X \rightarrow \Sigma_{2g-1}$. If σ is in the kernel, i.e., σ fixes the ramification points of f ,

If X is not hyperelliptic, we show that $\text{Aut}(X)$ permutes the hyperosculation points of the canonical embedding. □

3. *Moduli of Curves of Genus 4.* The hyperelliptic curves of genus 4 form an irreducible family of dimension 7. The nonhyperelliptic ones form an irreducible family of dimension 9. The subset of those having only one g_3^1 is an irreducible family of dimension 8.

Proof. Hyperelliptic elliptic curves of genus 4 are determined as two-fold coverings of $\mathbf{P}^1$, ramified at $0, 1, \infty$, and $2g - 1 = 7$ additional points, up to the action of a certain finite group. If X is a nonhyperelliptic curve of genus 4, then its canonical embedding in $\mathbf{P}^3$ is the complete intersection of a unique irreducible quadric surface and an irreducible cubic surface. Conversely, if X is a nonsingular curve in $\mathbf{P}^3$ which is a complete intersection of a quadric and a cubic surface, then X is a canonical curve of genus 4. There are only two isomorphism classes of irreducible quadric surfaces in $\mathbf{P}^3$, so the family of all nonhyperelliptic curves of

genus 4 is parametrized by two copies of an open set $U \subseteq \mathbf{P}^N$ with $N = 19$, because a form of degree 3 has 20 coefficients. Fixing a quadric surface $Q \subseteq \mathbf{P}^3$, there is a dominant morphism $U \rightarrow \mathfrak{M}_4$ defined by $F \mapsto Q \cap F$, so it remains to compute the dimension of a general fibre. Let F, F' be two irreducible cubic surfaces. Since the embedding is canonical, $Q \cap F$ and $Q \cap F'$ are isomorphic if and only if they differ by an automorphism of $\mathbf{P}^3$. $\square$

5. *Curves of Genus 5.* Assume X is not hyperelliptic.

- (a) The curves of genus 5 whose canonical model in $\mathbf{P}^4$ is a complete intersection $F_1.F_2.F_3$ form a family of dimension 12.
- (b) X has a g_3^1 if and only if it can be represented as a plane quintic with one node. These form an irreducible family of dimension 11.
- (c) In that case, the conics through the node cut out the canonical system (not counting the fixed points at the node). Mapping $\mathbf{P}^2 \rightarrow \mathbf{P}^4$ by this linear system of conics, show that the canonical curve X is contained in a cubic surface $V \subseteq \mathbf{P}^4$, with V isomorphic to $\mathbf{P}^2$ with one point blown up (II, Ex. 7.7). Furthermore, V is the union of all the trisecants of X corresponding to the g_3^1 (5.5.3), so V is contained in the intersection of all the quadric hypersurfaces containing X . Thus V and g_3^1 are unique.

Proof.

- (a)
- (b)
- (c)

$\square$

7. (a) Any automorphism of a curve of genus 3 is induced by an automorphism of $\mathbf{P}^2$ via the canonical embedding.
- (b) Assume $\text{char } k \neq 3$. If X is the curve given by

$$x^3y + y^3z + z^3x = 0,$$

the group $\text{Aut } X$ is the simple group of order 168, whose order is the maximum $84(g-1)$ allowed by (Ex. 2.5).

- (c) Most curves of genus 3 have no automorphisms except the identity.

Proof.

- (a)
- (b)
- (c)

$\square$